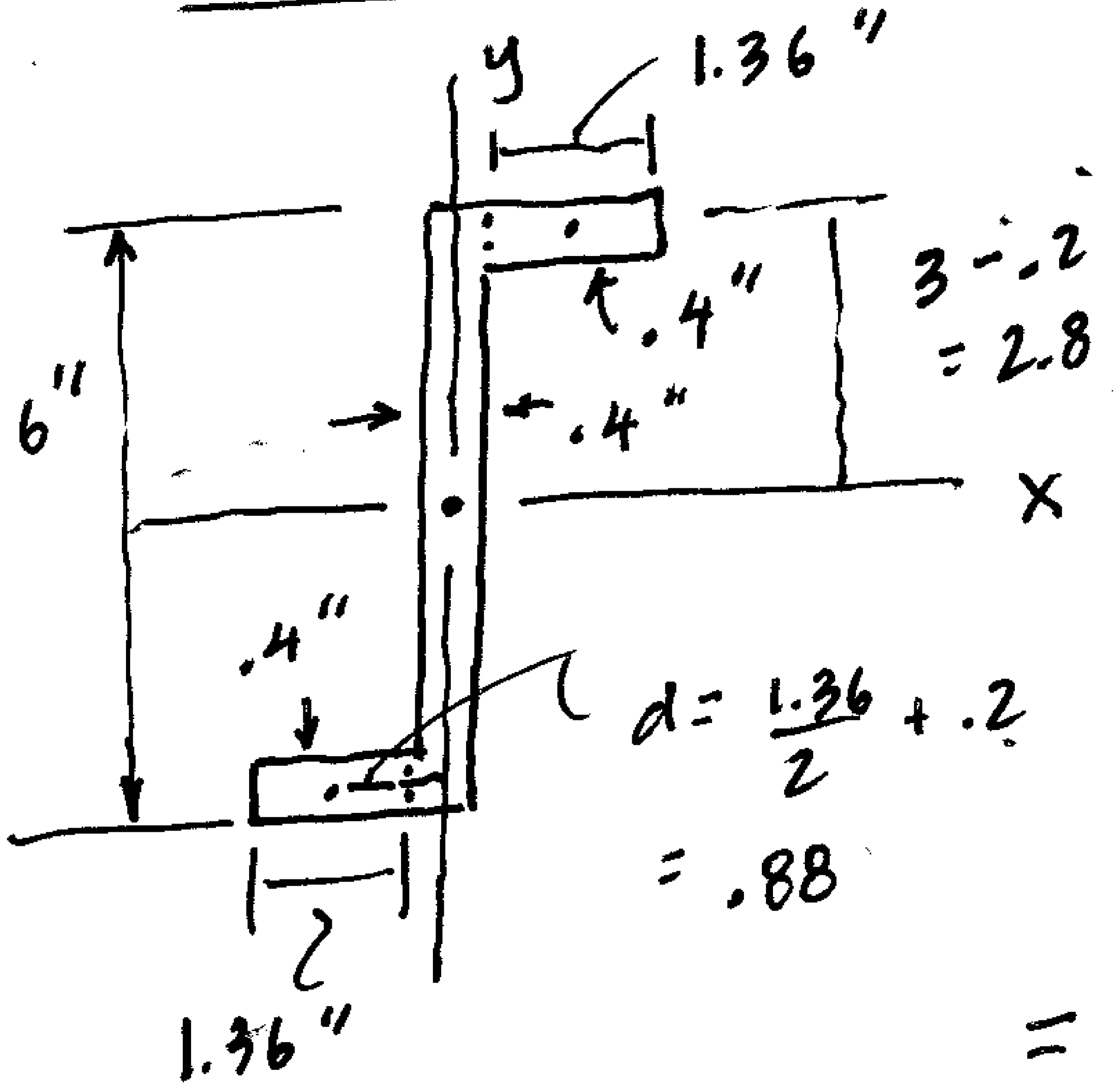
Example MC Problem:

- Integer θ_p
- Example of using MC to find $[I_u, I_{uv}]$ at given θ



$$I_x = \Sigma(\bar{I} + Ad^2)$$

$$= \frac{1}{12}(.4)(6)^3 + 2\left[\frac{1}{12}(1.36)(.4)^3 + (1.36)(.4)(2.8)^2\right]$$

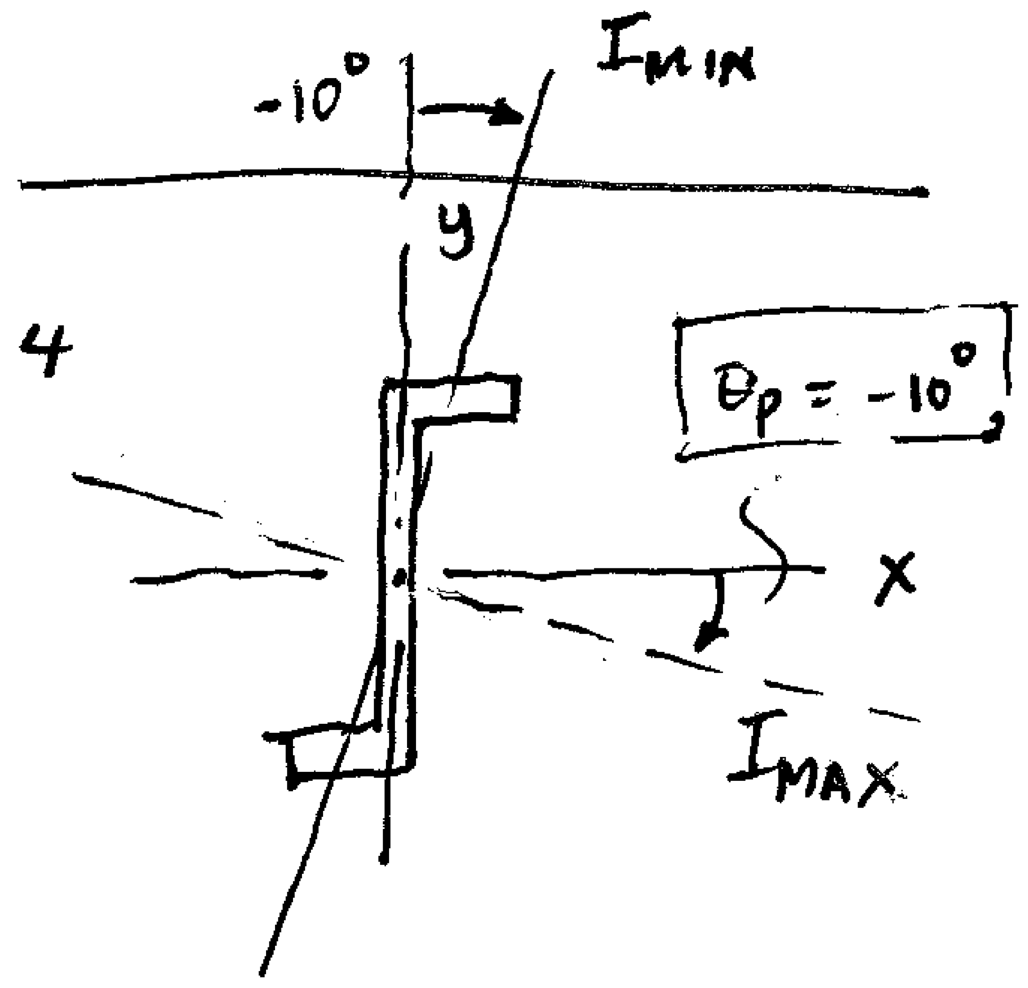
$$= 7.2 + 2(4.27) = \boxed{15.7 \text{ in}^4 = I_x}$$

$$I_y = \Sigma(\bar{I} + Ad^2) = \frac{1}{12}(6)(.4)^3 + 2\left[\frac{1}{12}(.4)(1.36)^3 + (1.36)(.4)(.88)^2\right]$$

$$I_y = .032 + 2(.505) = \boxed{1.04 \text{ in}^4 = I_y}$$

$$I_{xy} = \Sigma(\bar{I}_{xy} + A d_x d_y) = 0 + 2\left[0 + 1.36(.4)(.88)(2.8)\right]$$

$$= \boxed{2.68 \text{ in}^4 = I_{xy}}$$



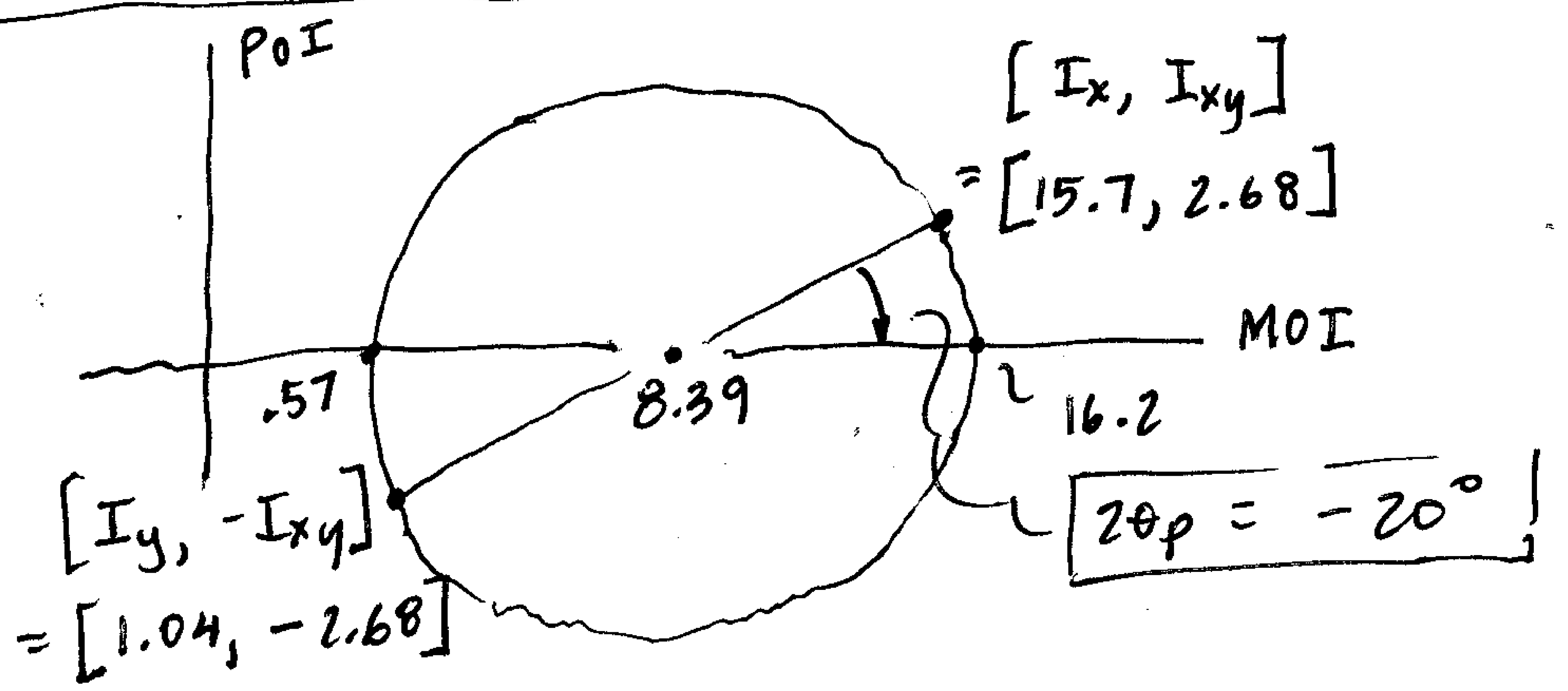
Set up MC:

$$I_{AVG} = \frac{15.7 + 1.04}{2} = 8.39 \text{ in}^4$$

$$\frac{I_x - I_y}{2} = \frac{15.7 - 1.04}{2} = 7.35 \text{ in}^4$$

$$R = \sqrt{7.35^2 + 2.68^2} = 7.82 \text{ in}^4$$

$$2\theta_p = \tan^{-1}\frac{-2.68}{7.35} = \boxed{-20° = 2\theta_p} \quad \boxed{\theta_p = -10°}$$



$$I_{MAX} = 8.39 + 7.82 = 16.2 \text{ in}^4$$

$$I_{MIN} = 8.39 - 7.82 = 0.57 \text{ in}^4$$